

almost exactly on the schedule that you would expect from Moore's Law.

These chips perform efficiently as well. The A7's Geekbench single central processing unit (CPU) score is 269 while for the A15 the score stands at 1734. This increase in score is mapped against multi-core performance as well as the graphical processing unit (GPU) performance.

Cost scaling with design scaling

If we were to estimate the amount of tape outs that occur per year and do any kind of matrix multiplication of tape outs per node against this, we will quickly run out of money in the industry. We as a company are currently doing a seven manometer (7nm) chip. The trend line is mapped well whether the numbers are exactly correct for any particular chip or not.

It becomes increasingly more expensive to do design as we move down these technology nodes. The design cost does not double, so the design cost per transistor continues to be cheaper.

The history of the EDA industry has been about increasing layers of abstraction. Back in the eighties, a tremendous amount of work was done at the transistor and SPICE (a general-purpose, open source analogue electronic circuit simulator software used in IC and board-level design to check the integrity of circuit designs and to predict circuit behaviour) level. In the late eighties and early nineties, the transition to RTL occurred. The transition to RTL enables us to design more quickly.

Understanding PAVE360

Siemens introduced a simple but revolutionary pre-silicon autonomous validation program, PAVE30, in May 2019 to accelerate the development of autonomous vehicle platforms. It provides an environment for multi-supplier collaboration across the automotive ecosystem for the development of automotive chips.

PAVE360 also extends digital twin simulation beyond processors to include automotive hardware and software sub-systems, full vehicle models, fusion of sensor data, traffic flows, and the simulation of smart cities through which self-driving cars will ultimately travel.

We have a sensing environment where we can model scenarios of real world data to where you can put traffic lights, traffic lanes, pedestrians, and other vehicles, all in place that allows you to run these various scenarios and then produce sensor data that can be virtualised and fed into the compute and analysis function. **EFY**

This article is based on a tech talk at VLSI 2022 by Joe Sawicki, Executive Vice President, Siemens Digital Industries Software. It has been transcribed and curated by Laveesh Kocher, an EFY tech enthusiast with a knack for open source exploration and research.

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